PedalEase: Converting Casual Riders into Loyal Annual Members

Introduction

PedalEase is an innovative bike-sharing platform serving major cities across the United States, offering flexible mobility options to urban travelers. The company's service caters to two distinct user segments: annual members—who benefit from unlimited rides, and casual riders—who use bikes on a pay-per-use basis. As PedalEase aims for sustained growth, the focus is shifting toward maximizing recurring revenue by converting casual riders into annual members. Understanding rider behavior, trip patterns, and user preferences is crucial for shaping effective strategies that bolster membership conversion and overall business performance.

Problem Statement

Despite casual riders generating a considerable portion of overall trips, annual members deliver higher lifetime value, revenue stability, and brand loyalty. PedalEase's challenge is to pinpoint actionable differences in usage patterns and behaviors between the two segments, identify high-potential casual riders for conversion, and devise targeted interventions that drive membership sign-ups. Achieving this requires deep analysis of ride data to:

- Quantify differences in ride frequency, trip duration, bike type preference, and location-specific demand between casual and member riders.
- Identify times, locations, and rider profiles where conversion opportunities are highest.
- Support decision-making for data-driven marketing and operational improvements.

Analysis Approach

The analysis is conducted using the most recent 12 months of ride data, and adopts the following structured approach:

- **KPI Breakdown:** Defining measurable key performance indicators such as ride length, ride count, bike type usage, and station diversity for each rider segment.
- **Behavioral Comparison:** Evaluating and visualizing differences in trip duration, frequency, day-of-week and time-of-day usage, and bike preferences between casual and member riders.
- **Location Analysis:** Ranking popular start and end stations; highlighting location-specific demand and station diversity to inform targeted campaigns.
- **Trend Analysis:** Reviewing monthly, weekly, and daily patterns to uncover seasonality and peak periods for both segments.

Dataset Overview

This analysis uses data from over 5 million PedalEase rides recorded in the last year, covering details like trip duration, rider type, bike usage, and station activity. With this data, we explore how casual and member riders behave differently, uncover patterns in when and where rides happen, and identify riders most likely to become members.

- ride_id : Unique identifier for each bike trip, applicable to both members and casual riders.
- rideable_type : Type of bike used for the ride. Can be one of categories: classic, docked, electric, etc.,
- member_casual" : "Indicates whether the rider is an annual member or a casual (non-member) rider.
- start_day_of_week : Day of the week on which the trip started.
- started_at : Date and time the trip started in YYYY-MM-DD HH:MM:SS format.
- ended_at : Date and time the trip ended in YYYY-MM-DD HH:MM:SS format.
- ride_length : Duration of the ride in hours, minutes, and seconds.
- "start_station_name" :"Name of the station where the bike was picked up.",
- "start_station_id" : "Unique identifier for the station where the bike was picked up.",

- "end_station_name" : "Name of the station where the bike was returned.",
- "end_station_id" : "Unique identifier for the station where the bike was returned.",
- "start_lat": "Latitude coordinate of the station where the bike was picked up.",
- "start_lng": "Longitude coordinate of the station where the bike was picked up.",
- "end_lat ": "Latitude coordinate of the station where the bike was returned.",
- "end_lng" : "Longitude coordinate of the station where the bike was returned."

Hypotheses:

1. Members have a significantly shorter average ride length than casual riders.

	member_casual	mean	median	count	percentage
0	casual	1334.896195	775.0	2343465	40.73
1	member	739.354196	526.0	3410683	59.27

- Conclusion: Confirmed. Members consistently ride for less time than casual riders.
- Recommendation: Highlight membership savings for frequent, short rides in marketing materials and signup prompts.

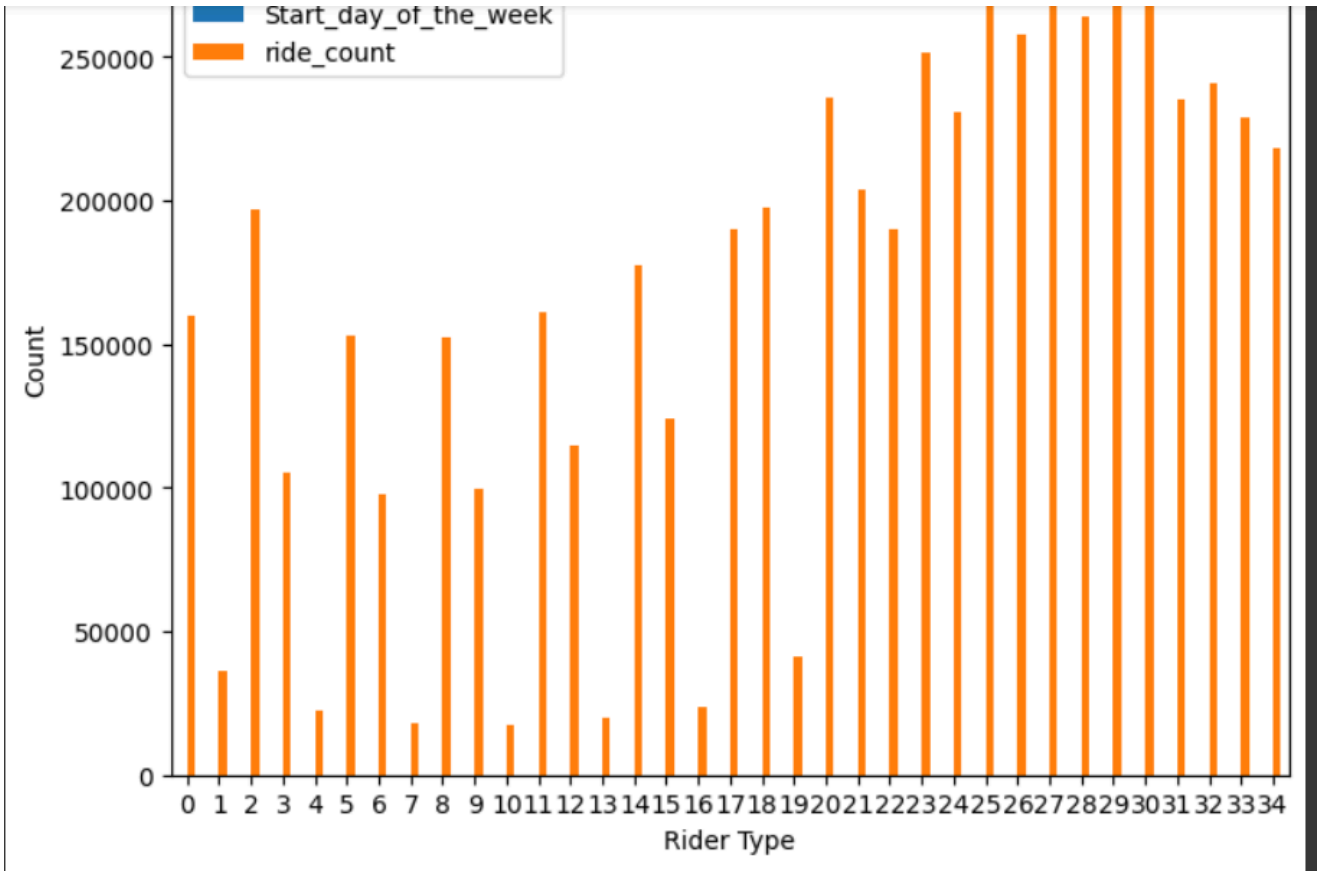
2. Ride frequency is higher for members than casual riders.

	member_casual	mean	median	count
0	casual	1.000014	1.0	2343487
1	member	1.000026	1.0	3410639

- Conclusion: Confirmed, though unique user tracking is limited. Members represent more overall trips and repeat usage.

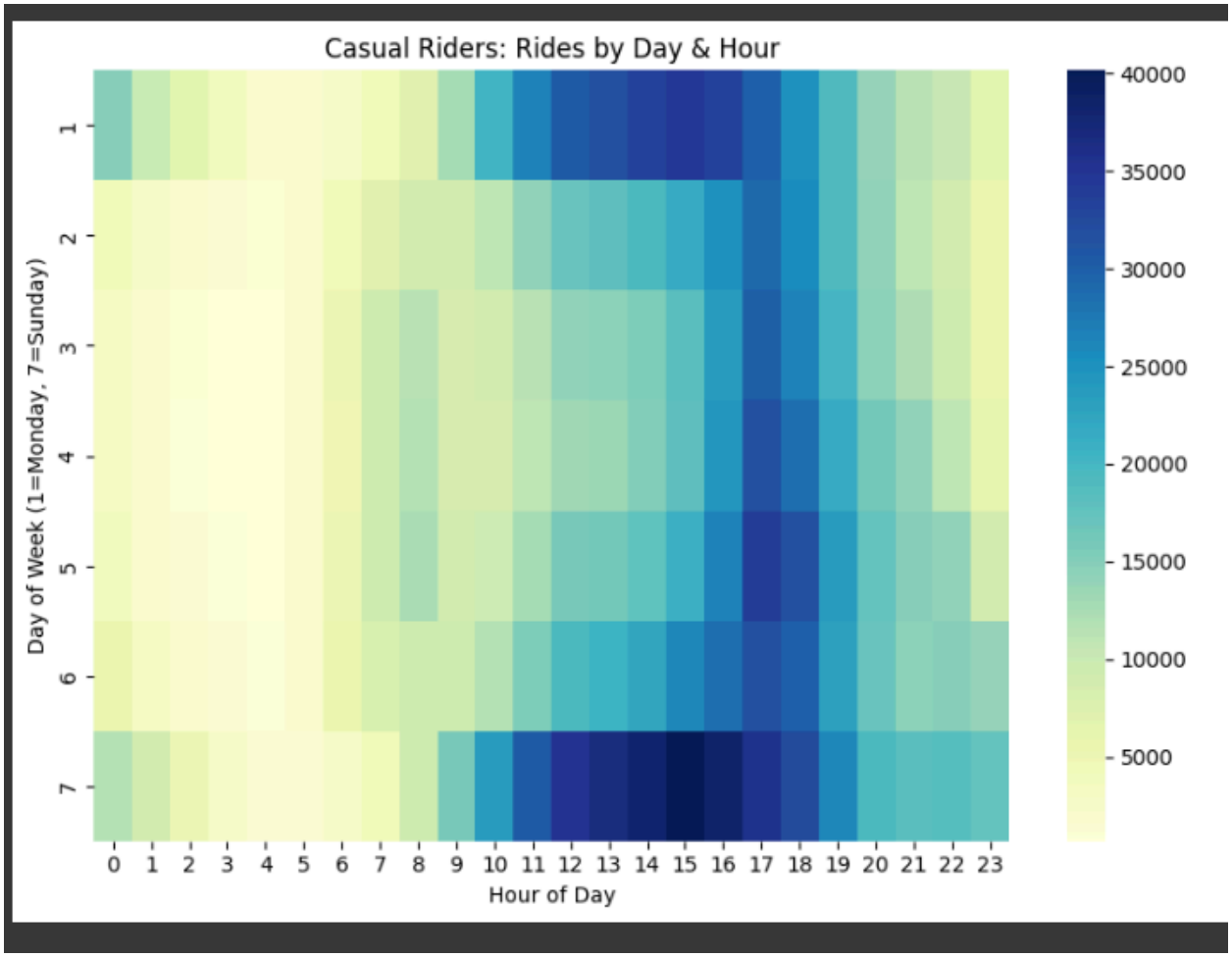
- Recommendation: Encourage casual riders to try multiple rides or multi-day passes to increase loyalty and demonstrate value.

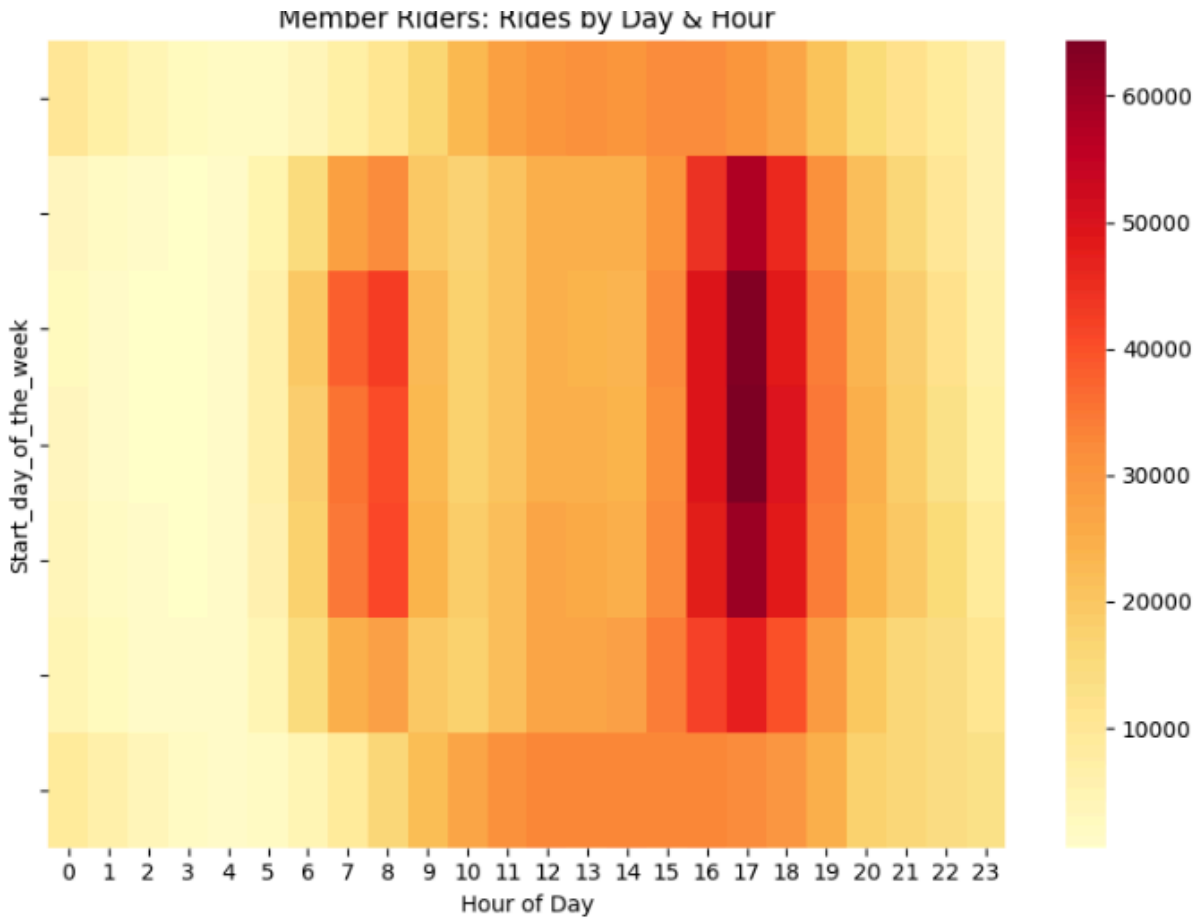
3. Casual riders prefer electric and docked bikes; members favor classic bikes.



- Conclusion: Confirmed. Casuals choose electric and docked bikes much more often than members, who mainly use classic bikes.
- Recommendation: Promote special rates or perks for electric/docked bikes to casuals and upsell members on expanded bike options.

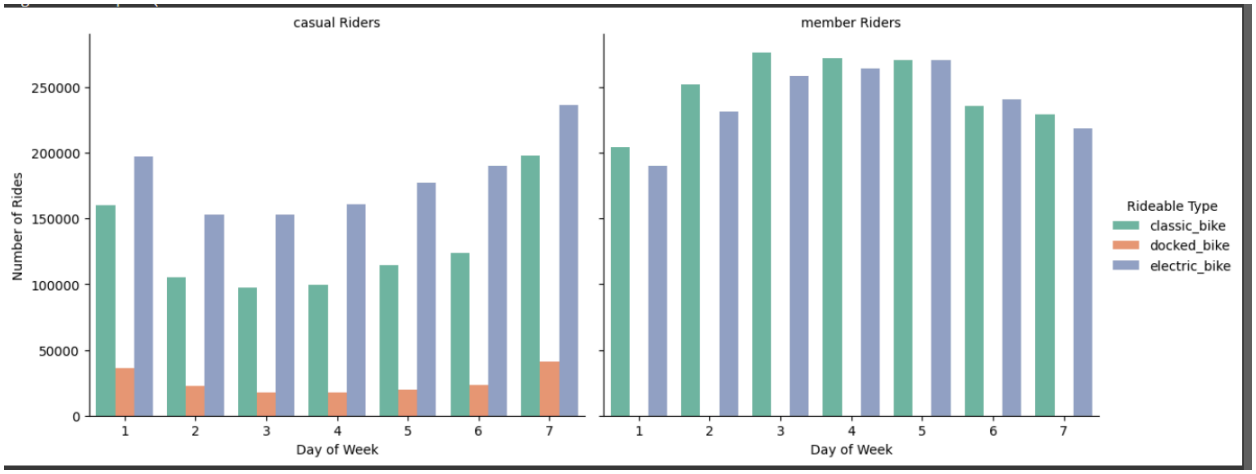
4. Casual riders' trips concentrate on weekends/evenings; members on weekday mornings/evenings.


Casual Riders: Rides by Day & Hour
Day of Week (1=Monday, 7=Sunday)
1
2
3
4
5
6
7
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
Hour of Day
40000
35000
30000
25000
20000
15000
10000
5000



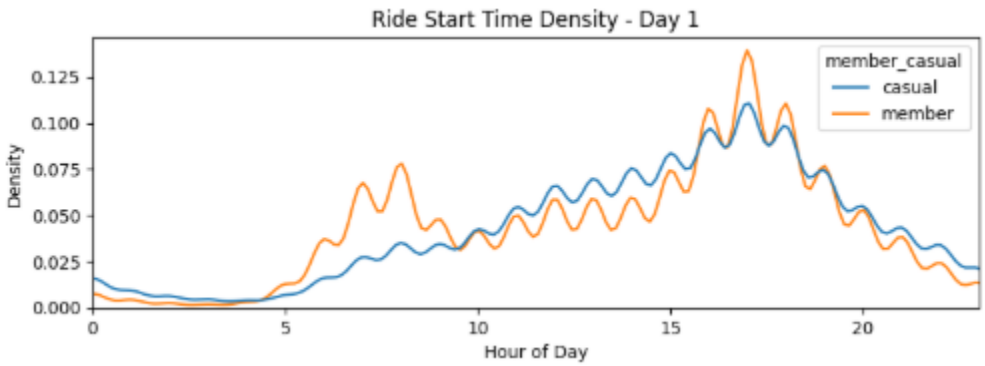
- Conclusion: Confirmed by temporal analysis. Members' ridership peaks during traditional commute times, while casual rider peaks occur on weekends and afternoons.
- Recommendation: Time campaigns for peak casual activity—weekend afternoons—and incentivize off-peak member use.

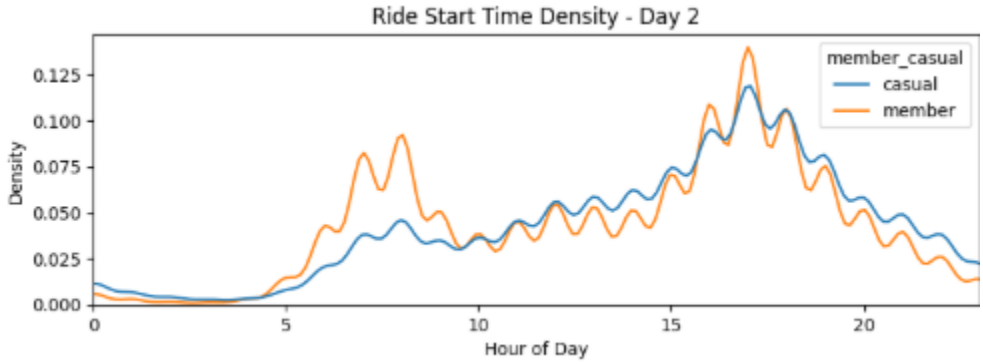
6. Rideable_type usage varies by weekday for casual and member riders (interaction effect).

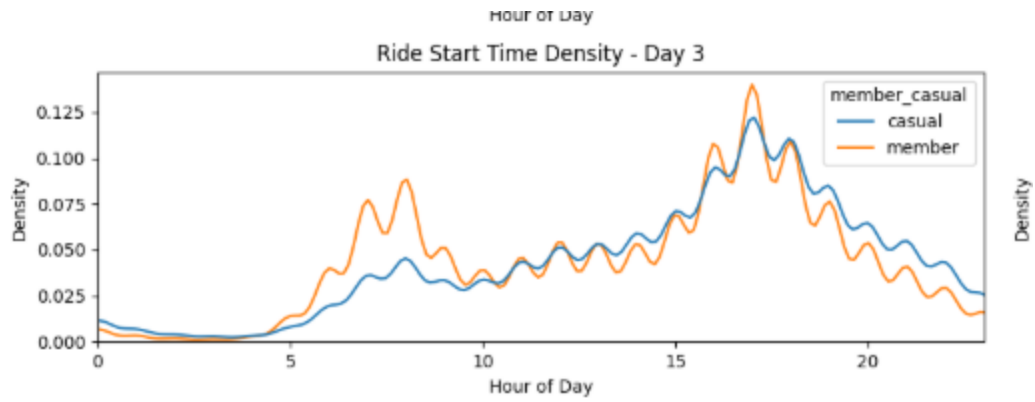


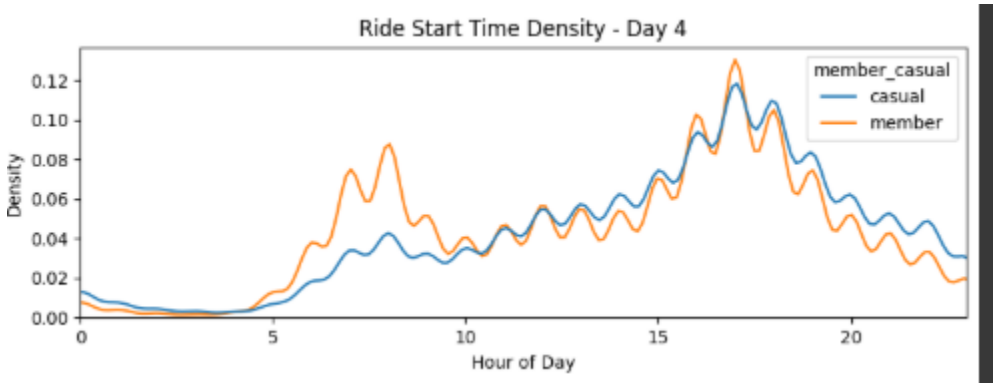
- Conclusion: Confirmed. Interaction plots show clear differences in bike choice by day and segment.
- Recommendation: Offer weekday-specific promotions matching casual preferences (e.g., electric bikes on weekends).

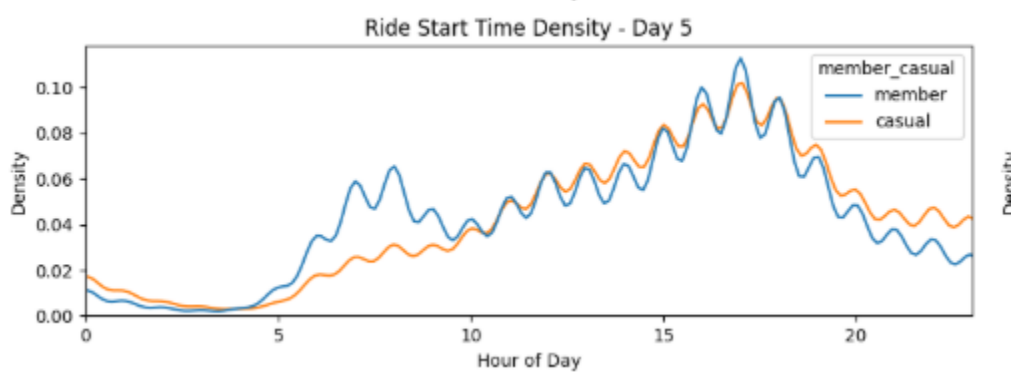
7. Peak ride times differ between casuals and members across weekdays.

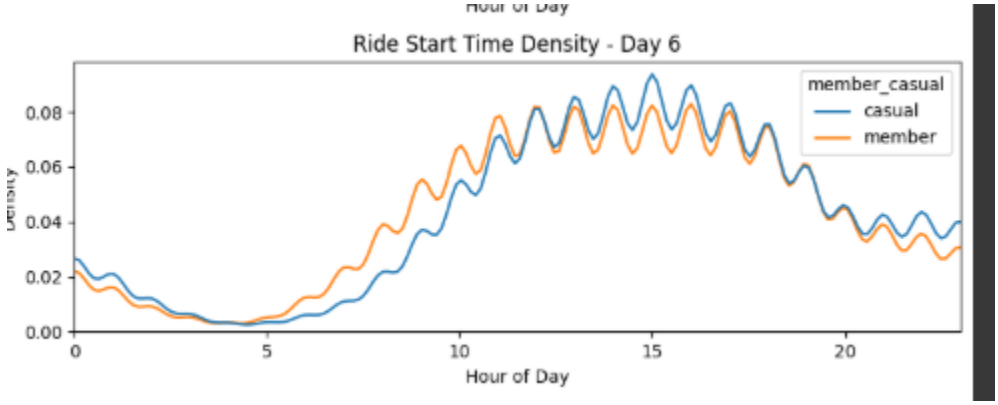


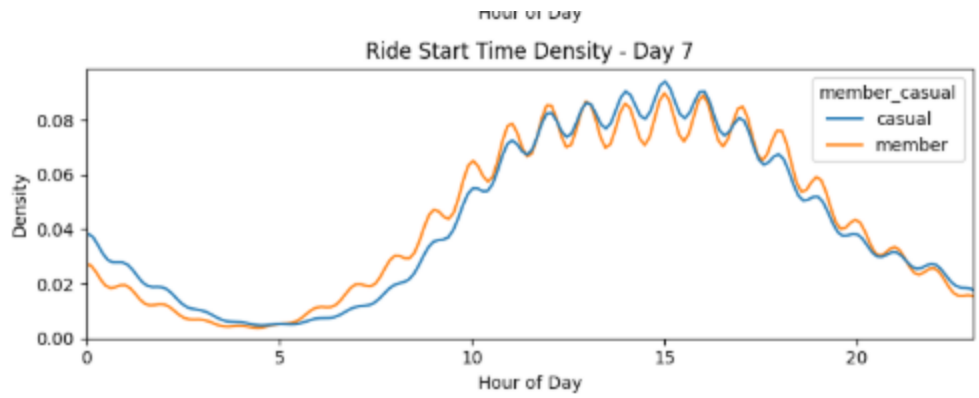



Ride Start Time Density - Day 3
member_casual
casual
member
Density
Hour of Day


Ride Start Time Density - Day 4
member_casual
casual
member
Density
Hour of Day


Ride Start Time Density - Day 5
member_casual
member
casual
Density
Hour of Day


Ride Start Time Density - Day 6
member_casual
casual
member
Hour of Day



- Conclusion: Confirmed. Casuals peak afternoons/weekends; members peak commute hours.
- Recommendation: Schedule push notifications for casuals during weekend afternoons and for members during week commute periods.

Key Observations & Findings

1. Trip Duration and Frequency

- Casual riders take fewer rides but have significantly longer average and median ride lengths per trip (mean ≈ 22 min, median ≈ 13 min) compared to members (mean ≈ 12 min, median ≈ 9 min).

- Members account for more total trips (≈ 59% of rides), with briefer, more routine travel lengths—reflecting commute and daily use behaviors.
- Average ride count per ride_id is essentially 1 for both groups, confirming lack of user-level grouping in raw data.

- Duplicate rides are rare; both segments have negligible repeat ride IDs.

2. Bike Type Usage Patterns

- Members strongly prefer classic bikes (≈ 51% usage vs. 38% for casual), while casual riders disproportionately use electric bikes (54% for casual vs. 49% for members).

- Docked bikes are used mainly by casual riders (8% share) and nearly never by members.

3. Peak Usage Days & Times

- Weekends (Day 7) show peak casual rider activity, outpacing member rides on Sundays.
- Members exhibit consistent ridership across weekdays, peaking modestly on workdays.

- Peak hours: Casual riders concentrate rides during late afternoons and weekends; members ride more during morning and evening commute windows.

4. Station Popularity & Location Demand
- Casual riders favor a mix of numbered/internal stations and some high-traffic public stations (e.g., Streeter Dr & Grand Ave). Internal station codes with high casual counts may indicate issues with data labeling or system exits worth review.

- Members cluster at key networked transit and commuter stations (University Ave & 57th St, Ellis Ave & 60th St), matching routine travel patterns.

- Station diversity: Casual riders use slightly more unique locations, suggesting more exploratory or leisure-based travel patterns than members, whose behavior is more concentrated.

5. Monthly & Geographical Trends
- Seasonal spikes: Casual ridership and ride length soar in spring and summer (May–August), then taper in winter—members remain steadier but also peak with favorable weather.

- Geographic distance: Average trip distances (≈ 2.18 km for casual, 2.12 km for members) are very similar for both segments, but casual rides remain slightly longer on average.

Temporal density plots confirm: Casual ridership peaks afternoons on weekends; members rally mornings/evenings on weekdays.

Business Takeaways & Recommendations

1. Longer casual rides mean higher propensity for conversion: Focus membership marketing on high-usage, longer-trip casual riders—show them the cost savings of unlimited rides.

2. Leverage electric bike popularity among casuals: Highlight exclusive member benefits for electric bike access, and run seasonal promotions to encourage trial-to-membership upgrades.

3. Target peak usage times and stations: Design campaign messaging and pop-ups for afternoons/weekends at stations with heavy casual draw; incorporate transit-oriented perks for member hotspots.

4. Station & trip insights point to tailored bundles: Develop flexible memberships for leisure/travel-focused casual riders, especially during summer spikes; promote commuter plans at member-dense stations.

5. Clean up station metadata: Review and resolve high-volume internal station codes to ensure true location analysis and improve campaign targeting.

6. Recognize routine vs. exploratory usage: Members' repeat, short-trip patterns suggest satisfaction with core service—casual riders' exploration signals need for educational messaging about full membership value (flexibility, convenience, savings).

7. Seasonal upswing drives opportunity: Time offers and conversion pushes to align with peak casual ridership months for optimal ROI.